Week 8: Multi-Agent System

Report

1. Agent Architecture Design

Agents (3 agents):

Agent	Role	Tools
Planner	Research Strategy Planner	QueryAnalysisTool
Retrieval	Information Retrieval Specialist	RAGSearchTool
Synthesis	Research Report Writer	SummarizerTool, FactCheckerTool

Workflow:

User Query → Planner → Conditional Routing → Retrieval (if needed) → Synthesis → Result

2. Implementation (CrewAI)

- **Shared State Management:**
  - ResearchState model with fields: query, complexity_score, needs_rag, strategy, retrieved_context, citations, final_answer
- **Conditional Routing:**
  - Planner decides needs_rag=true/false
  - If needs_rag=true and documents indexed → Retrieval → Synthesis
  - If needs_rag=false → Synthesis only
- **Human-in-the-Loop:**
  - Parameter human_in_the_loop=True
  - CrewAI Task with human_input=True

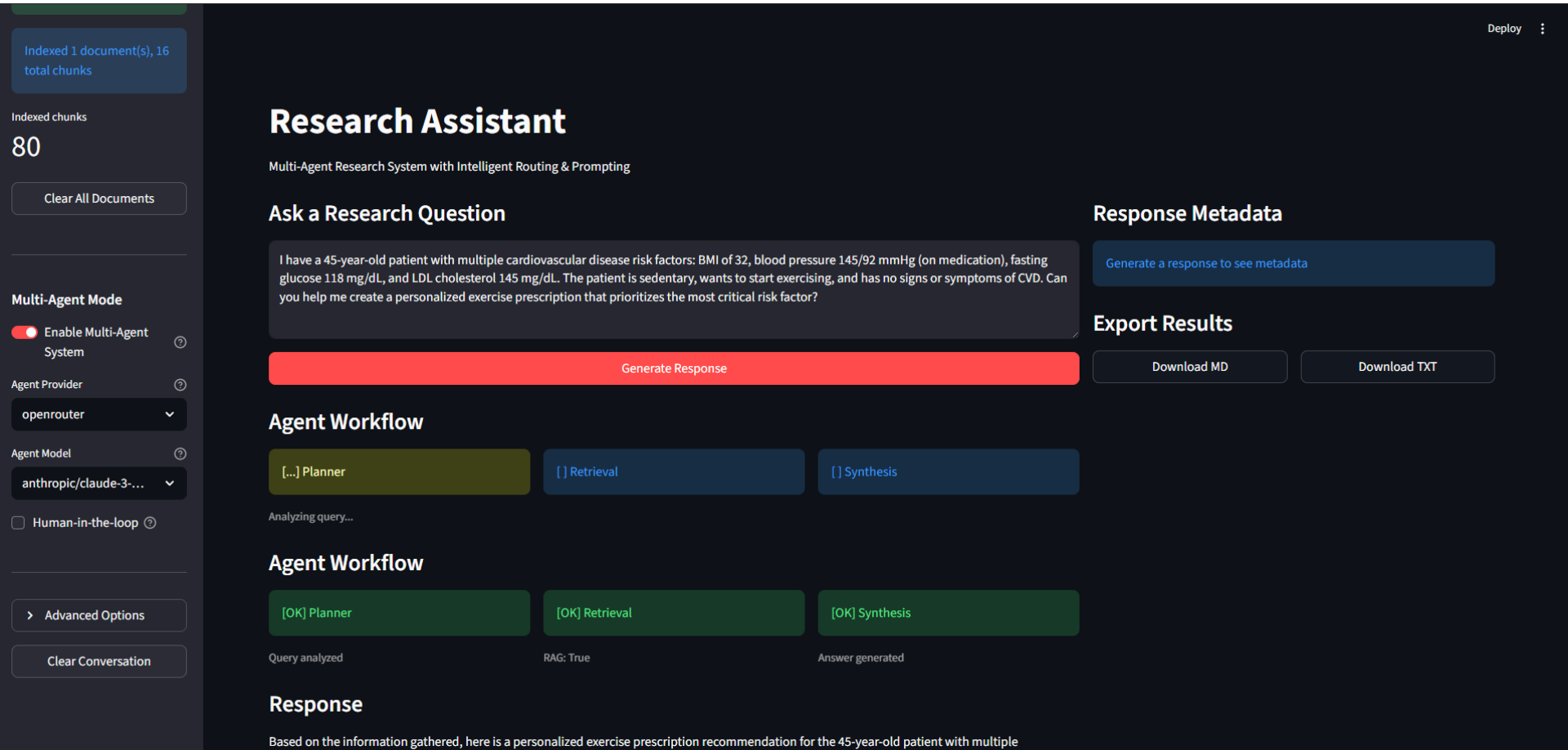
3. Tool Integration

Tool	Agent	Function
QueryAnalysisTool	Planner	Complexity 0-100, question type, needs_rag
RAGSearchTool	Retrieval	Semantic search, returns chunks with sources
SummarizerTool	Synthesis	Extracts key sentences
FactCheckerTool	Synthesis	Verifies claims against documents

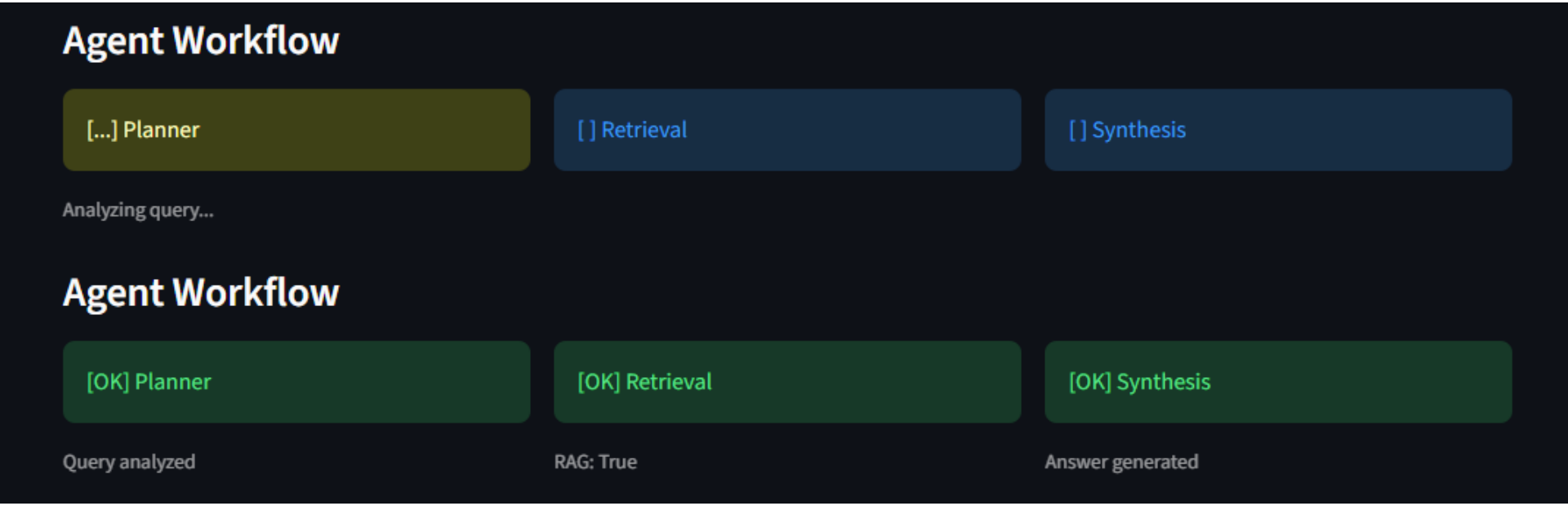
4. Complete System

UI additions:

- Multi-Agent Mode toggle in sidebar
- Provider/model selection for agents
- Agent step visualization (Planner → Retrieval → Synthesis)
- Export: Download MD, Download TXT



Agent Workflow in Action



5. Files Changed

File	Changes
src/agents/models.py	ResearchState
src/agents/definitions.py	Tools for Synthesis agent
src/agents/crew.py	Conditional routing, human_in_the_loop
src/tools/agent_tools.py	SummarizerTool, FactCheckerTool
src/ui/app.py	Multi-Agent Mode, visualization, export
README.md	Documentation section